

Week 1: Project Genesis and Architectural Framework to Continuous ASL Translation

Introduction and Project Scope: -

This development journal is an account of the preliminary stage of the building of a real time continuous American Sign Language (ASL) to English text translator. The main aim of the same project is to deal with the high communication barrier between the deaf and the hard-of-hearing community and the rest of the hearing community. Although more than one point five million people in the United States use ASL as their first language, the limited number of available real-time interpretation tools often frustrate communication in medical, educational, and social environments.

This project aims to design a software-purpose solution based on computer vision and deep learning. With the implementation of regular camera input instead of depending on special and limiting hardware such as sensor gloves, the resulting application will focus on accessibility, scalability and ease of use.

Architectural Development and Literature: -

In my early development of this project, I was planning to construct a stand-alone sign language recognition system. The method is usually based on a Convolutional Neural Network (CNN) to process images of hand shapes at rest and place them into relevant words.

But as I read the literature this week, it became apparent that a strictly spatial approach is inadequate. ASL is a very dynamic, spatiotemporal language; meaning is expressed not just by fixed configurations of hands, but by trajectory, velocity, and expressions, across time.

This epiphany required a significant change of direction in architecture. In a 2024 MDPI study of video-based sign language recognition, important quantitative data regarding this transition was obtained. The scientists proved the fact that a typical 3D-CNN (I3D) had a 65.89% classification accuracy. On the other hand, when a Hybrid CNN-LSTM architecture (that is, combining a spatial feature extractor with a temporal memory network) is used, accuracy rises to 84.65%. Moreover, a recent 2025 IEEE paper that employed a similar hybrid method had an average error of 92%. The results of this paper clearly show that only temporal modeling can result in a successful sign language recognition solution.

The Problems of Continuous Translation: -

The transition to Continuous Sign Language Translation (CSLT) and a step past isolated word classification presents two major technical challenges that I have been researching and sketching out solutions to over the course of this week:

The Boundary Problem (Co-articulation): In the normal use of signs, signers do not pause between different signs. The physical movement between the end of one sign and the beginning of the next can create so-called noise that a machine learning model can confuse with another gesture. The system should have the ability to read unsegmented video streams without definite boundary delimiters.

Syntactical and Grammatical fluctuation: ASL has a unique grammatical framework that is not directly comparable to English syntax. An example of this would be the English sentence "I am going to the store,"

which can be signed conceptually as "Store I go." A one-to-one translation (as it is also called glossing) would lead to discontinuous, unnatural texts.

Four-Stage Implementation Pipeline Proposed: -

To meet the spatiotemporal needs of ASL, as well as the nuances of nonstop translation, I have completed the following four-stage pipeline that will guide the remainder of my development:

Spatial Feature Extraction: The system will make use of MediaPipe Holistic offered by Google to process the frames of a live video. This architecture will identify dense, 3D spatial keypoints on the hands, face and body posture of the user to produce the physical condition of the signer at one millisecond interval.

Temporal Sequence Modeling: The extracted keypoints are going to be fed into an Encoder-Decoder Long Short-Term Memory (LSTM) net. This element will follow the transformation of the spatial data through time, which will essentially be the dynamic movement of the signs.

Connectionist Temporal Classification (CTC): CTC Loss will be used to optimize the model so that the problem of the boundary of continuous signature can be solved. In this method, the neural network is trained to match an unsegmented sequence of input frames to an output sequence of words and does not require manually segmented training data.

Integration of Natural Language Processing: The raw sequence of ASL glosses generated by the LSTM will be sent to a Large Language Model (LLM) to address the syntax difference between ASL and English. The LLM will put the sequence into context and rearrange the output as grammatically valid English sentences.